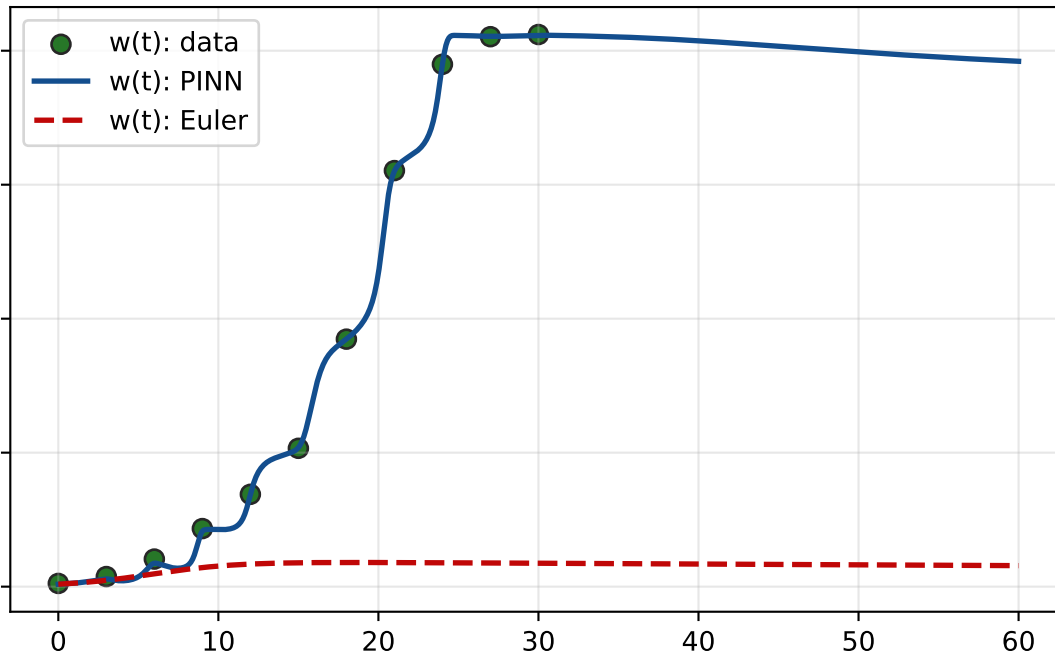
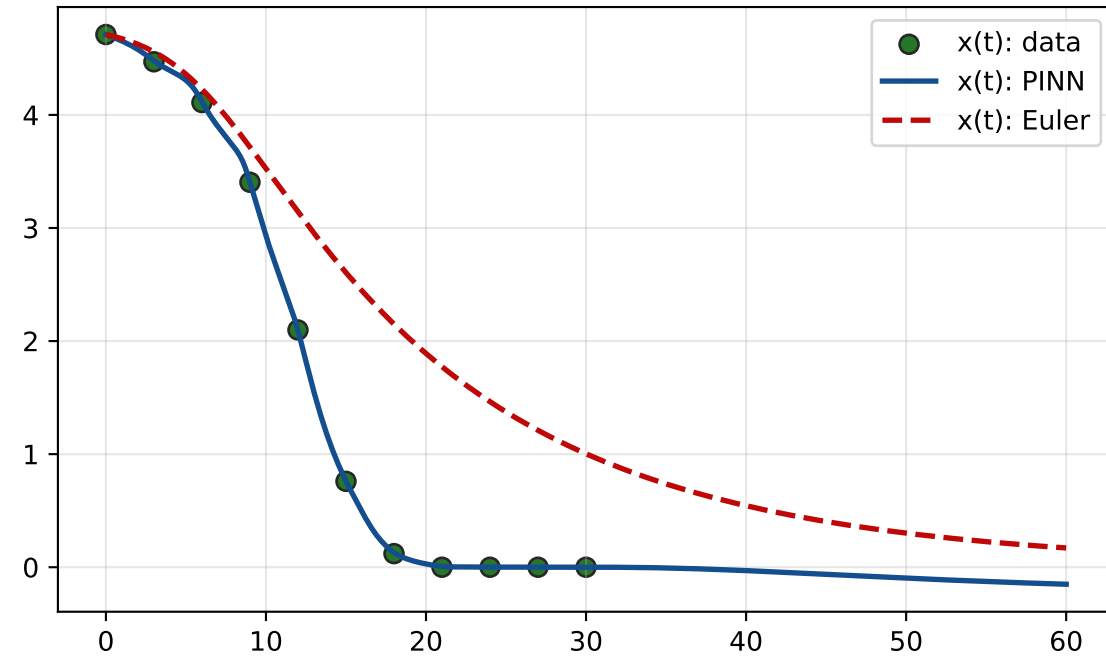


# E5 | PINN vs Euler

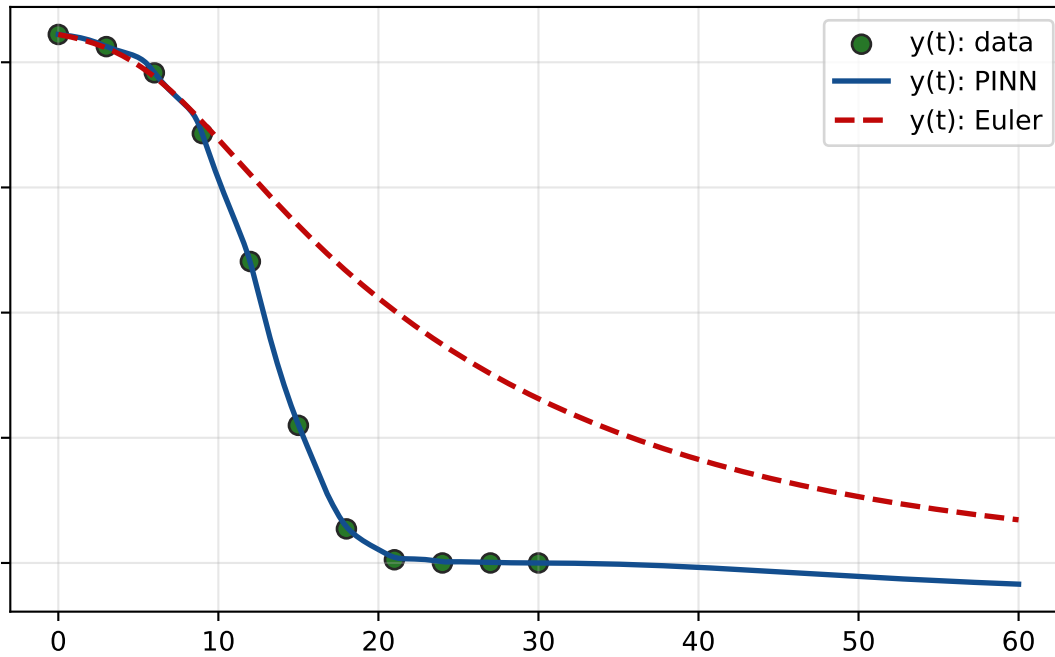
w(t): Biomass



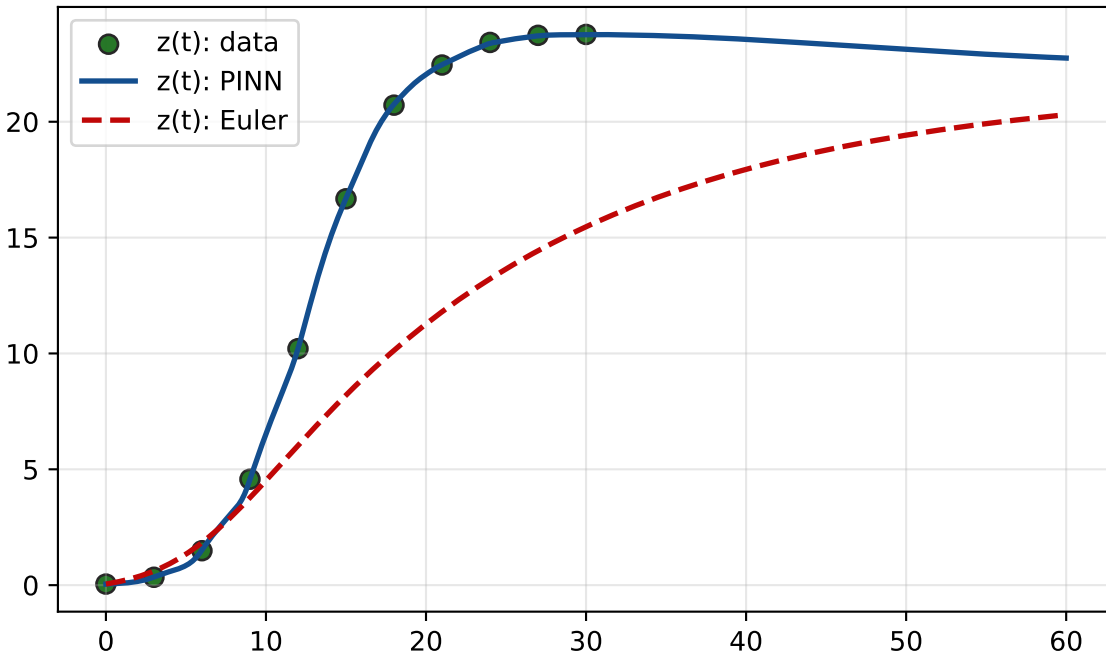
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E5

R2=0.4293, R2\_adj=0.3368, RSS=2.650e+03, AIC=1.946e+02

cond(JTJ)=3.45e+08, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 3.354302e-01 | 2.151653e+02 | 4.355794e+02 | -4.352439e+02 | 4.359148e+02 | 9.987643e-01 |
| p2    | 1.000000e+00 | 7.745553e+01 | 1.568005e+02 | -1.558005e+02 | 1.578005e+02 | 9.897666e-01 |
| p3    | 1.000000e-02 | 2.173262e+02 | 4.399539e+02 | -4.399439e+02 | 4.399639e+02 | 9.999635e-01 |
| p5    | 1.783622e-01 | 1.227462e+00 | 2.484867e+00 | -2.306505e+00 | 2.663229e+00 | 8.852344e-01 |
| p7    | 1.333168e-01 | 8.821682e-01 | 1.785856e+00 | -1.652539e+00 | 1.919173e+00 | 8.806774e-01 |
| p9    | 6.555277e-02 | 4.340081e-01 | 8.786034e-01 | -8.130506e-01 | 9.441562e-01 | 8.807428e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 3.650695e-02 |
| x0 (fixed)     | 4.710870e+00 |
| y0 (fixed)     | 4.222028e+01 |
| z0 (fixed)     | 4.653331e-02 |
| W0 (fixed)     | 0.000000e+00 |